The Likelihood Ratio Test

Copyright © Barry Van Veen 2014

Feel free to pass this ebook around the web... but please do not modify any of its contents. Thanks!

AllSignalProcessing.com

Key Concepts

1) The likelihood ratio is a test for binary, simple hypotheses.

    a) Binary: a test of two hypotheses, H_0 vs H_1.

    b) Simple: the complete probability density functions of the data are known under both hypotheses.

2) A binary hypotheses test involves comparing a function $T(x)$ of the data x to a threshold and deciding H_1 is true if $T(x)$ exceeds the threshold and H_0 is true otherwise.

3) Neyman-Pearson Lemma: The likelihood ratio $T(x) = \frac{f_{H_1}(x)}{f_{H_0}(x)}$ is the test that results in the maximum probability of detection for any specified probability of false alarm.

   a) $f_{H_1}(x)$ and $f_{H_0}(x)$ are the probability density functions under H_1 and H_0, respectively, evaluated at the observed data x.

   b) Probability of false alarm is the probability of deciding H_1 when H_0 is true.

   c) Probability of detection is the probability of deciding H_1 when H_1 is true.

d) The likelihood $f_{H_i}(x)$ is the probability density function evaluated at the observed data x.

e) The likelihood $f_{H_i}(x)$ is proportional to the probability we measure data close to x assuming H_i is true.

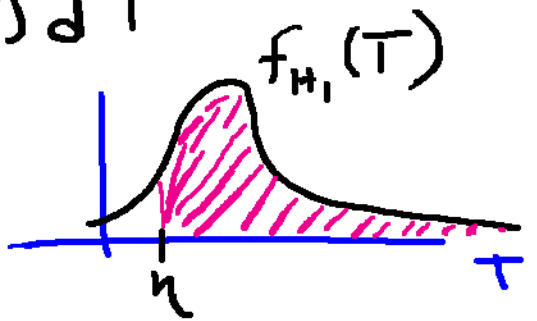
4) The receiver operating characteristic (ROC) for the likelihood ratio test is closer to the upper left corner than any other test.

Video Screenshots

Binary hypothesis test: $T(\underline{x}) \underset{H_0}{\overset{H_1}{\gtrless}} \eta$

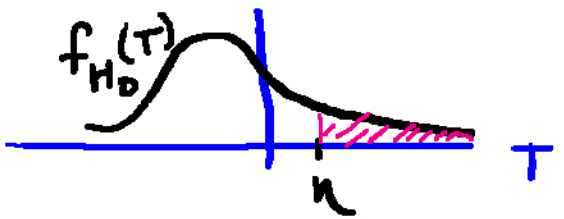
probability of detection

$$\text{Prob}\{T(\underline{x}) > \eta \mid H_1 \text{ true}\} = \int_{\eta}^{\infty} f_{H_1}(T)\, dT$$

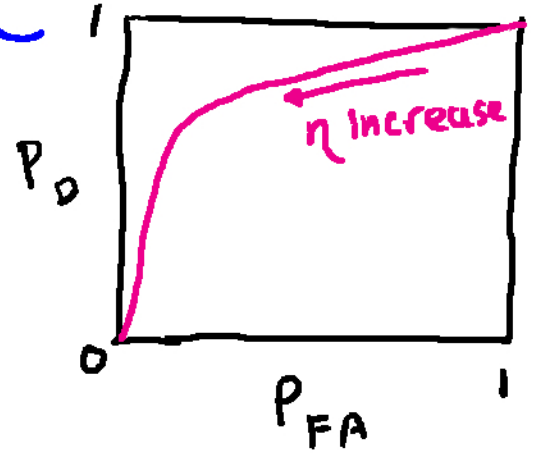


probability of false alarm

$$\text{Prob}\{T(\underline{x}) > \eta \mid H_0 \text{ true}\} = \int_{\eta}^{\infty} f_{H_0}(T)\, dT$$



ROC



Find <u>good</u> $T(\underline{x})$?

For simple hypotheses ($f_{H_1}(\underline{x})$ and $f_{H_0}(\underline{x})$ known completely)

Neyman-Pearson Lemma (observe $\underline{x}$)

The test that maximizes P_D for any given P_{FA}:

$$L(\underline{x}) = \frac{f_{H_1}(\underline{x})}{f_{H_0}(\underline{x})} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

"likelihood ratio"

where γ satisfies $P_{FA} = \int_{\gamma}^{\infty} f_{H_0}(L)\, dL.$

Likelihood: $\text{Prob}\{|\underline{x} - \underline{\alpha}| < \varepsilon\} = \int_{\underline{x}-\varepsilon}^{\underline{x}+\varepsilon} f(x)\, dx \approx f(\underline{x})\, \Sigma$ likelihood



$$L(\underline{x}) = \frac{\text{likelihood observe } \underline{x} \text{ given } H_1}{\text{likelihood observe } \underline{x} \text{ given } H_0}$$

Example: $H_0: x[n] = w[n], \; n = 0,1,\ldots N-1$ "noise"

$H_1: x[n] = s[n] + w[n], \; n = 0,1,\ldots,N-1$ "signal plus noise"

$s[n]$ is known, $w[n] \sim N(0,\sigma^2)$ with known σ^2 (Simple!)

Let $\underline{s} = [s[0] \cdots s[N-1]]^T$, $\underline{w} = [w[0] \cdots w[N-1]]^T$

$H_0: \underline{x} = \underline{w}$ $\quad f_{H_0}(\underline{x}) \sim N(0, \sigma^2 \underline{\underline{I}}) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left\{-\frac{1}{2\sigma^2}\underline{x}^T\underline{x}\right\}$

$H_1: \underline{x} = \underline{s} + \underline{w}$ $\quad f_{H_1}(\underline{x}) \sim N(\underline{s}, \sigma^2 \underline{\underline{I}}) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left\{-\frac{1}{2\sigma^2}(\underline{x}-\underline{s})^T(\underline{x}-\underline{s})\right\}$

Likelihood Ratio

$$L(\underline{x}) = \frac{f_{H_1}(\underline{x})}{f_{H_0}(\underline{x})} = \exp\left\{-\frac{1}{2\sigma^2}\left[(\underline{x}-\underline{s})^T(\underline{x}-\underline{s}) - \underline{x}^T\underline{x}\right]\right\}$$

$$L(\underline{x}) > \gamma \Rightarrow L'(\underline{x}) = \ln(L(\underline{x})) > \ln(\gamma) \qquad L'(\underline{x}) = -\frac{1}{2\sigma^2}\left(\underline{s}^T\underline{s} - 2\underline{s}^T\underline{x}\right)$$

11:06

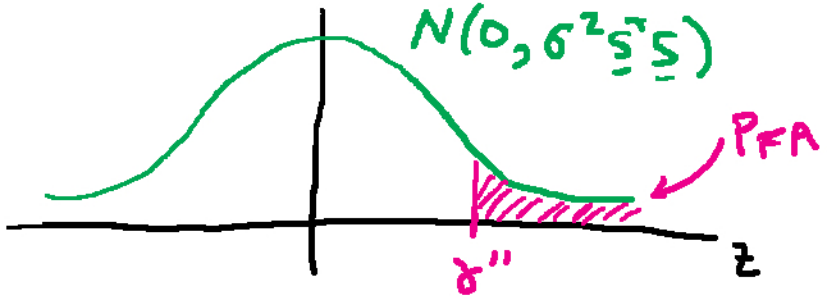
Absorb σ^2, $\underline{s}^T\underline{s}$ (known) into threshold

$$L''(\underline{x}) = \left(L'(\underline{x})\,2\sigma^2 + \underline{s}^T\underline{s}\right)/2 \quad = \quad \underline{s}^T\underline{x} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma'' \quad = \quad (\gamma' 2\sigma^2 + \underline{s}^T\underline{s})/2$$

Note $\underline{s}^T\underline{x} = \sum_{n=0}^{N-1} s[n]\,x[n]$ "matched filter"

To find γ'' for desired P_{FA}: need pdf of $\underline{s}^T\underline{x}$ under H_0

$H_0: \underline{x} \sim N(\underline{0}, \sigma^2 \underline{I}) \Rightarrow \underline{s}^T\underline{x} \sim N(0, \sigma^2 \underline{s}^T\underline{s})$ $\qquad z = \underline{s}^T\underline{x}$



$$P_{FA} = \int_{\gamma''}^{\infty} \frac{1}{(2\pi\sigma^2\underline{s}^T\underline{s})^{1/2}} \exp\left\{-\frac{1}{2\sigma^2\underline{s}^T\underline{s}}\, z^2\right\} dz$$

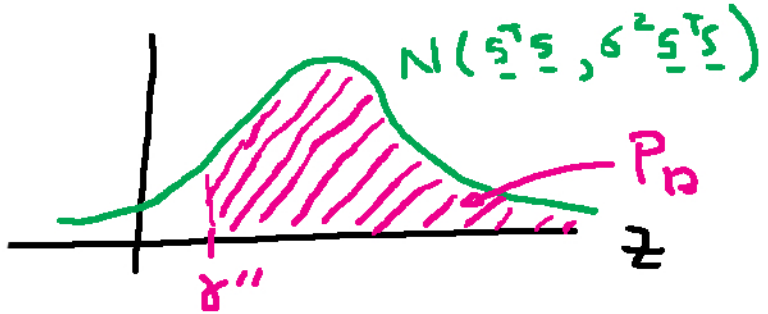
γ'' is easily found with a computer

Performance characterization

14:07

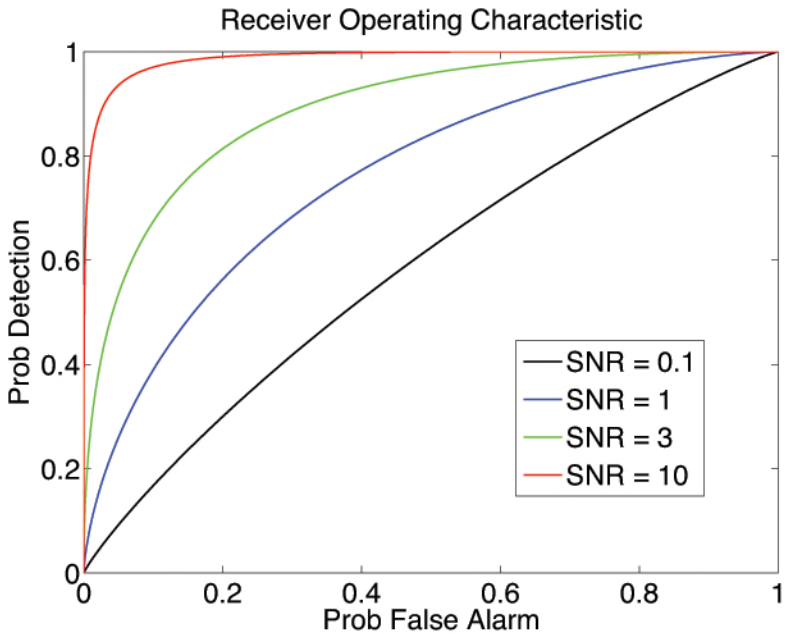
$$H_1: z = \underline{s}^T \underline{x} \sim N(\underline{s}^T \underline{s}, \sigma^2 \underline{s}^T \underline{s})$$

$$P_D = \int_{\gamma''}^{\infty} \frac{1}{(2\pi\sigma^2 \underline{s}^T \underline{s})^{1/2}} \exp\left\{-\frac{1}{2\underline{s}^T \underline{s}\sigma^2}(z - \underline{s}^T \underline{s})^2\right\} dz$$



For fixed P_{FA}, P_D depends on signal-to-noise ratio

$$SNR = \frac{\underline{s}^T \underline{s}}{\sigma^2} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} s^2[n]$$



Test performance ↑ as SNR ↑

For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction